

Abhishek Singh

CS undergrad at KIIT with hands-on experience building and deploying full-stack AI applications using Python, FastAPI, React, and Docker.
[gmail](#) | [portfolio](#) | [linkedin](#) | [github](#) | [itch.io](#)

EDUCATION

Kalinga Institute of Industrial Technology (KIIT)
Bachelor of Technology in Computer Science and Engineering

Bhubaneswar, India
2023 – Present | CGPA: 8.48

PROJECTS

Talk with CSV | *React, FastAPI, Groq AI (Llama 3.3), Python, Pandas, Matplotlib* | [Live](#) | [GitHub](#) 2026

- Built a full-stack AI-powered analytics platform enabling natural-language queries over uploaded CSV datasets
- Designed a FastAPI backend using Groq LLM APIs to generate insights, tables, and visualization instructions
- Implemented automated chart generation and deployed the application using React, Render, and modern REST APIs

Disease Risk Predictor | *Python, Scikit-learn, Streamlit, Docker* | [Live](#) | [GitHub](#) 2026

- Built an end-to-end ML web app predicting Diabetes and Heart Disease risk from real medical datasets
- Applied SMOTE for class imbalance, engineered interaction features, and ran GridSearchCV across 405 model fits
- Compared Logistic Regression, Random Forest, Gradient Boosting, and SVM; achieved 92.44% accuracy on heart disease
- Saved trained models and scalers as .pkl files; applied consistent StandardScaler transformation at inference
- Containerized with Docker and deployed live on Render

Network Intrusion Detection System | *Python, Scikit-learn, TensorFlow, Pandas, NumPy* | [GitHub](#) 2026

- Developed an anomaly-based intrusion detection system using the KDD Cup 1999 cybersecurity dataset
- Implemented and compared Isolation Forest and Autoencoder models for unsupervised attack detection
- Trained Autoencoders on normal network traffic and used reconstruction error for anomaly identification
- Achieved 95.81% accuracy, 98.37% recall, and 91.87% F1-score, significantly outperforming Isolation Forest

Game Development – IIT Guwahati Game Jam (GameDeon) | *Godot, GDScript* | [itch.io](#) 2025

- Secured **7th position among 50 teams** nationwide; invited to IIT Guwahati for final showcase
- Built and shipped a complete game as part of a 2-member team under strict time constraints

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C, C++, JavaScript

Data Science & ML: Pandas, NumPy, Scikit-learn, TensorFlow, EDA, Feature Engineering, Model Evaluation, SMOTE, Hyperparameter Tuning

Algorithms: Random Forest, Gradient Boosting, Logistic Regression, SVM, Isolation Forest, Autoencoders, TF-IDF, Cosine Similarity

Visualization: Matplotlib, Seaborn

Web & APIs: React, FastAPI, REST APIs, HTML, CSS

Databases & Tools: SQL, MariaDB, SQLite, Git, GitHub, Docker, Linux, Jupyter Notebook, Streamlit

CERTIFICATIONS

IBM Data Science Professional Certificate Coursera
Courses: What is Data Science, Tools for Data Science, Data Science Methodology, Python for Data Science

Google: Foundations: Data, Data, Everywhere Coursera
Explored data-driven decision-making, data collection, analysis, and interpretation

ADDITIONAL

Competitive Programming

Regularly practice DSA on LeetCode and CodeChef (Python, C++); focus on arrays, linked lists, graphs, and DP